



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:14 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:16 PM**Time taken** 1 min 49 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int count_frequency(int *nums, int start, int end, int target) {
5     int count = 0;
6     for (int i = start; i <= end; i++) {
7         if (nums[i] == target) {
8             count++;
9         }
10    }
11    return count;
12 }
13
14 int get_majority(int *nums, int start, int end) {
15     if (start == end) {
16         return nums[start];
17     }
18
19     int mid = start + (end - start) / 2;
20     int left_majority = get_majority(nums, start, mid);
21     int right_majority = get_majority(nums, mid + 1, end);
22
23     if (left_majority == right_majority) {
24         return left_majority;
25     }
26
27     int left_count = count_frequency(nums, start, end, left_majority);
28     int right_count = count_frequency(nums, start, end, right_majority);
29
30     return (left_count > right_count) ? left_majority : right_majority;
31 }
32
33 int main() {
34     int n;
```

```
35 if (scanf("%d", &n) != 1) {
36     return 0;
37 }
38
39 int *nums = (int *)malloc(n * sizeof(int));
40 if (nums == NULL) {
41     return 1;
42 }
43
44 for (int i = 0; i < n; i++) {
45     scanf("%d", &nums[i]);
46 }
47
48 int result = get_majority(nums, 0, n - 1);
49 printf("%d\n", result);
50
51 free(nums);
52 return 0;
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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